SST-74: Defect-Bound Chirality Probes and Near-Zero Modes

A Translation Note from Impurity Spectroscopy in a Chiral Kagome Superconductor

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Abstract

We formulate a narrow Swirl-String Theory (SST) translation of a recent impurity-spectroscopy result in the kagome superconductor CsV_3Sb_5. The source paper reports two central features: first, a defect-localized resonance with a strongly mirror-broken spatial pattern interpreted as evidence for hidden chiral order; second, a robust zero-bias peak near a vacancy, discussed as a defect-bound near-zero mode associated with a sign-changing order-parameter boundary. The present note does not import the condensed-matter ontology literally. Instead, it extracts the transferable structural content and rewrites it in SST terms. The main translatable claim is that a localized defect can reveal hidden handed order of the host medium through a symmetry-broken localized response, and that a sign-changing branch field near a defect can support a near-zero localized mode. We state explicitly which parts translate cleanly, which remain analogical only, and which are currently unknown in SST. A minimal effective SST operator is proposed for numerical testing.

1 Purpose and scope

This note is intentionally narrow. It is not a claim that Kondo physics, Yu–Shiba–Rusinov (YSR) states, Bogoliubov quasiparticles, or Majorana zero modes are already present in canonical SST. The goal is more modest:

1. identify the structurally transferable content of the source paper,
2. rewrite that content in SST-compatible language,
3. state explicitly where the translation stops, and
4. propose a minimal SST defect-mode ansatz suitable for falsifiable numerical work.

The translation criterion used throughout is strict:

> A paper element is called *transferred* only if it can be rewritten as an SST-level field, defect, operator, or observable without silently importing orthodox condensed-matter ontology.

2 Source-paper kernel

The source paper studies impurity-induced local electronic states in the kagome superconductor CsV_3Sb_5. Its two central observational claims may be summarized as follows:

(a) Cr dopants produce a localized resonance whose spatial pattern breaks all in-plane mirror symmetries, interpreted as evidence for hidden chiral order in the host phase.

(b) V vacancies produce a robust zero-bias peak inside the superconducting gap, interpreted as a defect-bound near-zero mode consistent with a sign-changing order-parameter boundary.

For SST, the transferable core is therefore not the specific condensed-matter vocabulary but the following narrower mechanism:

$$\boxed{\text{A localized defect can reveal hidden handed order of the host medium through a symmetry-broken localized resp}} \tag{1}$$

A second transferable kernel is:

$$\boxed{\text{A sign-changing branch field near a defect can support a localized near-zero mode.}} \tag{2}$$

3 SST translation dictionary

We separate the translation into three categories.

3.1 Transferred

1. **Defect as probe of hidden order.** A localized defect in a handed background can act as a spectroscopic probe of the background handedness.

2. **Mirror-broken local response as chirality diagnostic.** If the defect couples to a pseudoscalar or parity-odd structure of the host medium, the localized response need not preserve mirror symmetry.

3. **Near-zero defect mode from branch reversal.** A sign-changing or branch-changing order field around a defect can produce a localized low-frequency mode.

3.2 Analogical only

1. **Kondo screening.** There is currently no canonical SST object corresponding to a Fermi-sea screening cloud around a magnetic impurity.

2. **Majorana mode language.** A strict Bogoliubov–de Gennes particle-hole Majorana interpretation is not currently part of SST.

3. **Topological superconducting surface state.** This phrase remains specific to band topology and superconducting order, not directly to canonical SST.

3.3 Unknown / not yet translatable

1. Which precise SST field should replace the sign-changing superconducting order parameter.

2. Which microscopic SST operator should replace the defect–chirality coupling of the source paper.

3. Whether the appropriate SST near-zero mode should be interpreted as torsional, swirl-clock, or another branch excitation.

4 Minimal SST fields

To express the translated mechanism, introduce the following coarse-grained fields.

First, a handedness field

$$\chi(\mathbf{x}) \in \{-1, +1\}, \tag{3}$$

which distinguishes two local chirality sectors of the medium.

Second, a branch or sign field

$$\sigma(\mathbf{x}) \in \{-1, +1\}, \tag{4}$$

which represents a local branch assignment of an order field that may reverse across a defect-centered boundary.

Third, a localized defect potential

$$V_{\mathrm{def}}(\mathbf{x}) = V_0\, f(\mathbf{x} - \mathbf{x}_0), \tag{5}$$

with center $\mathbf{x}_0$ and spatial scale ℓ.

A natural SST coarse-grained pseudoscalar associated with local handedness is the helicity-density-like quantity

$$h(\mathbf{x}) = \mathbf{u}(\mathbf{x}) \cdot \big(\nabla \times \mathbf{u}(\mathbf{x})\big), \tag{6}$$

where $\mathbf{u}$ is an effective medium velocity field. At phenomenological level one may let

$$\chi(\mathbf{x}) = \mathrm{sgn}\big(h(\mathbf{x})\big), \tag{7}$$

although this identification is not yet canonical.

5 Minimal effective operator

A minimal translated SST operator that captures both chirality-sensitive mirror breaking and defect-bound near-zero modes is

$$\Big[-\partial_t^2 + c_T^2 \nabla^2 - m_0^2 - m_1^2\, \sigma(\mathbf{x}) + \lambda_\chi\, \chi(\mathbf{x})\, \hat{\mathbf{n}} \cdot (\nabla \times) + V_{\mathrm{def}}(\mathbf{x})\Big]\, \psi = 0. \tag{8}$$

Here:

- c_T is an effective propagation speed for the relevant localized mode sector,
- m_0 is a baseline mode gap,
- $m_1^2 \sigma(\mathbf{x})$ is a branch-sensitive local mass term,
- λ_χ measures the strength of chirality-sensitive coupling,
- $\hat{\mathbf{n}}$ is a preferred local orientation set by the background structure,
- V_{def} is the defect-localizing perturbation.

Equation (8) is *not* derived from the source paper. It is a translation ansatz that captures the transferable operator-level structure.

6 Mirror-broken localized response

Suppose the background has nonzero handedness,

$$\chi(\mathbf{x}) \neq 0, \tag{9}$$

and that the defect couples to the parity-odd term

$$\lambda_\chi \, \chi(\mathbf{x}) \, \hat{\mathbf{n}} \cdot (\nabla \times). \tag{10}$$

Then the local mode intensity

$$I(\mathbf{x}; \mathbf{x}_0) = |\psi(\mathbf{x}; \mathbf{x}_0)|^2 \tag{11}$$

need not respect the mirror symmetries that would otherwise be expected from the surrounding geometry. In other words, localized defect spectroscopy can become a chirality diagnostic:

$$\chi \neq 0 \wedge \lambda_\chi \neq 0 \quad \Longrightarrow \quad I(\mathbf{x}) \text{ may be mirror-broken.} \tag{12}$$

This is the clean SST translation of the source paper's first central motif.

7 Near-zero mode from branch reversal

The second motif is the appearance of a robust near-zero localized mode when an order field changes sign near a defect. The structurally standard one-dimensional prototype is

$$\left[-\partial_x^2 + m^2(x)\right] \psi(x) = \omega^2 \psi(x), \tag{13}$$

with an odd mass profile

$$m(-x) = -m(x). \tag{14}$$

If the effective local gap reverses sign across a defect-centered interface, Equation (13) can support a mode with small ω localized near the interface.

In translated SST language, the sign-changing field is represented by $\sigma(\mathbf{x})$. The defect-centered radial prototype is

$$\sigma(r) = \begin{cases} -1, & r < r_0, \\ +1, & r > r_0. \end{cases} \tag{15}$$

Substituting this into Equation (8) gives a direct SST test model for a defect-bound near-zero mode.

8 What remains unknown

The translation remains incomplete in three important ways.

8.1 Unknown replacement for the superconducting sign field

The source paper uses a sign-changing superconducting order parameter. In SST, the correct counterpart is not yet fixed. Plausible candidates are:

$$Q(\mathbf{x}) = \chi(\mathbf{x}), \qquad Q(\mathbf{x}) = \Theta(\mathbf{x}) - \Theta_0, \qquad Q(\mathbf{x}) = S_t^{\circlearrowleft}(\mathbf{x}) - S_t^{\circlearrowleft}{}_0, \tag{16}$$

but none of these identifications is presently canonical.

8.2 Unknown microscopic coupling

The paper motivates a coupling between defect-localized response and hidden chirality. SST does not yet possess a derived microscopic operator for that coupling. The term

$$\lambda_\chi \, \chi(\mathbf{x}) \, \hat{\mathbf{n}} \cdot (\nabla \times) \tag{17}$$

is therefore only an effective phenomenological proxy.

8.3 Unknown mode identity

Even if a near-zero bound state is found in Equation (8), its interpretation is not yet determined. It could be:

(a) a torsional bound mode,

(b) a swirl-clock branch mode,

(c) a mixed chirality–branch excitation,

(d) or a more complicated defect-pinned collective mode.

9 Falsifiable numerical program

A direct numerical test can be built around Equation (8) in two spatial dimensions. Let

$$V_{\mathrm{def}}(r) = V_0 \exp\left(-\frac{r^2}{\ell^2}\right), \tag{18}$$

and define trial profiles

$$\chi(x,y) = \chi_0 \tanh\left(\frac{x}{w_\chi}\right), \qquad \sigma(x,y) = \tanh\left(\frac{x}{w_\sigma}\right). \tag{19}$$

Then the following predictions are sharp:

$$\chi_0 = 0 \Longrightarrow \text{the defect pattern should remain more mirror-symmetric,} \tag{20}$$

$$\chi_0 \neq 0 \Longrightarrow \text{the defect pattern may become mirror-broken,} \tag{21}$$

$$\sigma \text{ sign reversal} \Longrightarrow \text{a localized low-frequency mode becomes more likely.} \tag{22}$$

A practical observable is the local density

$$I(x,y) = |\psi(x,y)|^2, \tag{23}$$

compared across parameter scans in $(\lambda_\chi, m_1, V_0, \ell, w_\chi, w_\sigma)$.

10 Relation to current SST ontology

This note fits naturally with the current SST direction in which bosons are rings or simple strings and the photon is treated as a torsional pulse rather than a static knot. In that setting, a defect-bound near-zero mode is plausibly more naturally interpreted as a localized torsional or branch excitation than as a matter-knot excitation. That interpretation remains tentative, but it is ontologically closer to the present SST picture than a literal import of condensed-matter quasiparticles.

11 Main conclusion

The source paper does not provide direct SST mathematics in the strict sense. It does, however, provide a strong and useful structural lesson:

$$\boxed{\text{defects can diagnose hidden handed order, and sign-changing branch structure can trap near-zero modes.}} \tag{24}$$

That lesson translates well into SST and motivates a concrete effective operator, Equation (8), for numerical exploration. The translation is therefore partial but substantial:

$$\boxed{\text{philosophical transfer: strong}}, \quad \boxed{\text{mathematical transfer: moderate and structural}}, \quad \boxed{\text{literal ontology trans}} \tag{25}$$

Child-level analogy. A tiny bump on a spinning trampoline can reveal which hidden way the whole fabric prefers to twist, and if the twist changes sign across the bump, a small trapped wiggle can live there.

References

[1] Y. Tu, Z. Zhang, W. Lu, T. Han, R. Lv, Z. Wang, Z. Zhou, X. Hou, N. Hao, Z. Wang, X. Chen, and L. Shan, Symmetry-Broken Kondo Screening and Zero-Energy Mode in the Kagome Superconductor CsV_3Sb_5, arXiv preprint arXiv:2502.20733, 2025. DOI: `https://doi.org/10.48550/arXiv.2502.20733`.

[2] Y. Tu, Z. Zhang, W. Lu, Y. Xiao, T. Han, R. Lv, Z. Wang, Z. Zhou, X. Hou, N. Hao, X. Chen, and L. Shan, Symmetry-broken Kondo screening and zero-energy mode in a kagome superconductor, *Nature Physics*, published online 24 March 2026. DOI: `https://doi.org/10.1038/s41567-026-03223-5`.

[3] M. H. Christensen, T. Birol, B. M. Andersen, and R. M. Fernandes, Loop currents in AV_3Sb_5 kagome metals: Multipolar and toroidal magnetic orders, *Physical Review B* **106**, 144504 (2022). DOI: `https://doi.org/10.1103/PhysRevB.106.144504`.

[4] S. C. Holbæk, M. H. Christensen, A. Kreisel, and B. M. Andersen, Unconventional superconductivity protected from disorder on the kagome lattice, *Physical Review B* **108**, 144508 (2023). DOI: `https://doi.org/10.1103/PhysRevB.108.144508`.

[5] V. Madhavan, W. Chen, M. F. Crommie, N. S. Wingreen, and H. C. Manoharan, Tunneling into a Single Magnetic Atom: Spectroscopic Evidence of the Kondo Resonance, *Science* **280**, 567–569 (1998). DOI: `https://doi.org/10.1126/science.280.5363.567`.